

Guilherme Salles

US Citizen | Brazilian Citizen | EU Citizen
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EDUCATION

Carnegie Mellon University

Bachelor of Science in Mathematical Sciences, GPA: 3.9/4.0

Intended Major in Computer Science

Selected Coursework: Introduction to Mathematical Finance, Multidimensional Calculus, Principles of Imperative Computation, Matrices and Linear Transformations, Fundamentals of Programming and Computer Science

Pittsburgh, PA

May 2029

American Heritage Schools – Broward Campus

Diploma

Completed Coursework: Machine Learning & AI, Differential Equations, Linear Algebra

Plantation, FL

May 2025

SKILLS

Programming Languages and Tools: Python, C, LaTeX, Linux, Github, NumPy, Pandas, Scikit-learn, TensorFlow

Languages: English, Portuguese

EXPERIENCE

Undergraduate Researcher

Mathematical Sciences Department, CMU

May 2026 — July 2026

Pittsburgh, PA

- Experimented with songbirds, a theoretical language used by Raymond Smullyan in *To Mock a Mockingbird* to explore the theory of computation, under the guidance of professor Richard Statman.
- Created intermediary proofs to show that it is not possible to find a fixed-point combinator (Sage bird) using songbirds B , I , and M , establishing the inability to construct songbirds T and K in a restricted environment.

AI Software Engineer Intern

Starwood Property Trust

May 2025 — June 2025

Miami Beach, FL

- Using Python and the OpenAI library, trained and evaluated a custom GPT-based classifier on a large collection of PDF-extracted text to identify legal documents, achieving 98% accuracy on an internal test data set.

Teaching Assistant and Test Writer

Mu Alpha Theta Honor Society, American Heritage Schools

Plantation, FL

August 2022 — July 2025

- Organized 50+ classes, 15+ workshops, and wrote 15+ math competition tests in Geometry, Algebra 2, Precalculus, Calculus, and Linear Algebra.

PROJECT EXPERIENCE

Brief.ly – Application, Hackathon Submission

November 2025

NOVA Hackathon, Scotty Labs

Pittsburgh, PA

- Developed a newsletter service, with a team of four, that converts scraped news articles into structured summaries, personalized to specified user interests.
- Created the GPT-based summarization and integrated a multi-step pipeline spanning web scraping, summarization, and automated email delivery.

Tetris AI – AI Model

December 2024

American Heritage Schools - Broward Campus

Plantation, FL

- Designed and trained a Tetris agent in Python using a genetic algorithm, clearing 100,000+ lines in a single run.

HONORS & AWARDS

CMIMC Integration Bee 1st Place

March 2025

44 "Top 5" Awards in Mu Alpha Theta Competitions

January 2022 — July 2025

University of Central Florida High School Programming Competition, Top 5

March 2025

Three-time AIME Qualifier (Best Score: 7)

October 2022 — February 2025

ACTIVITIES

Carnegie Mellon Informatics and Mathematics Competition, Integration Bee Writer

August 2025 — Present

Carnegie Mellon University Math Club

August 2025 — Present

Quant Club

August 2025 — Present